

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

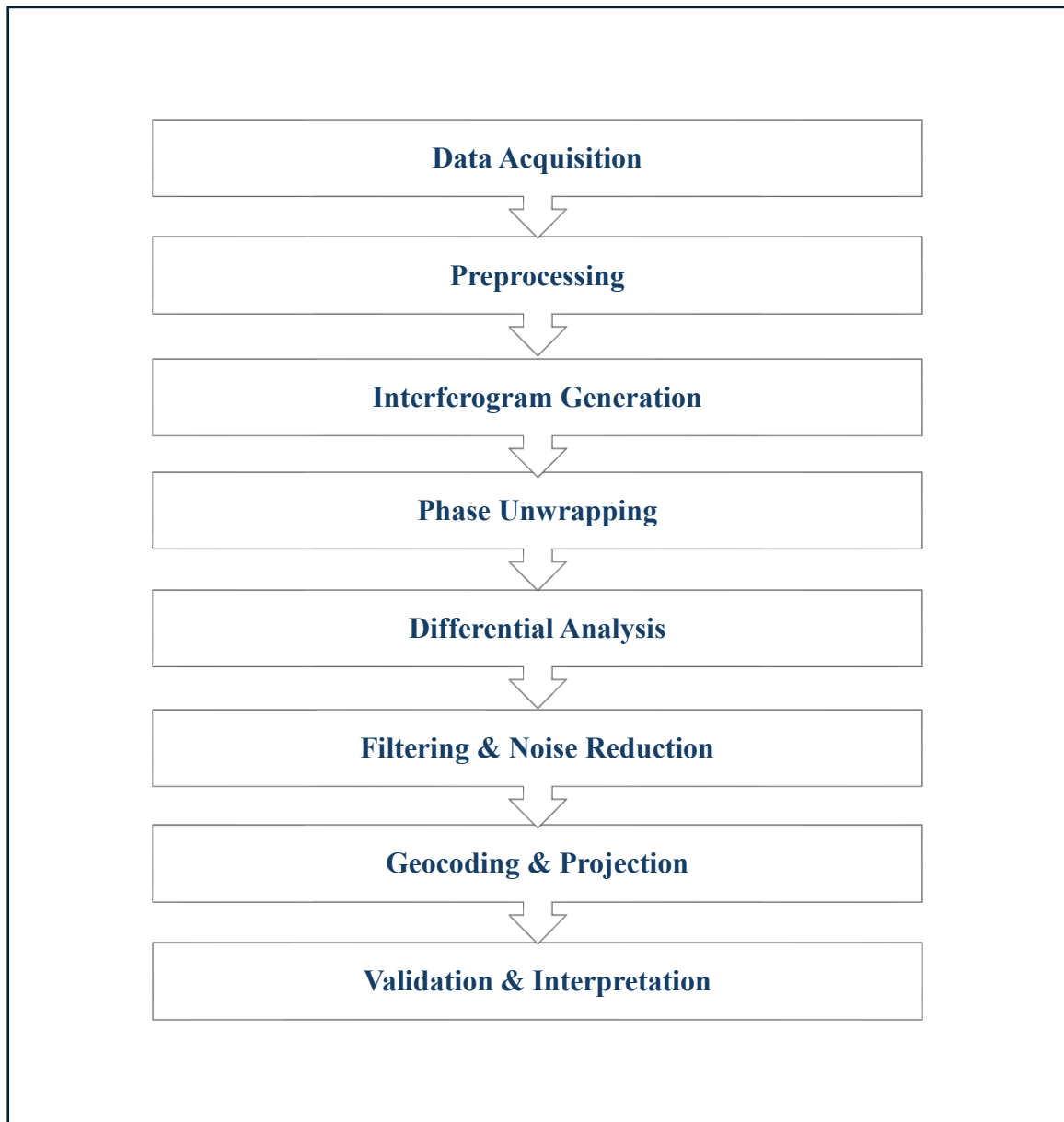
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 1039+200

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

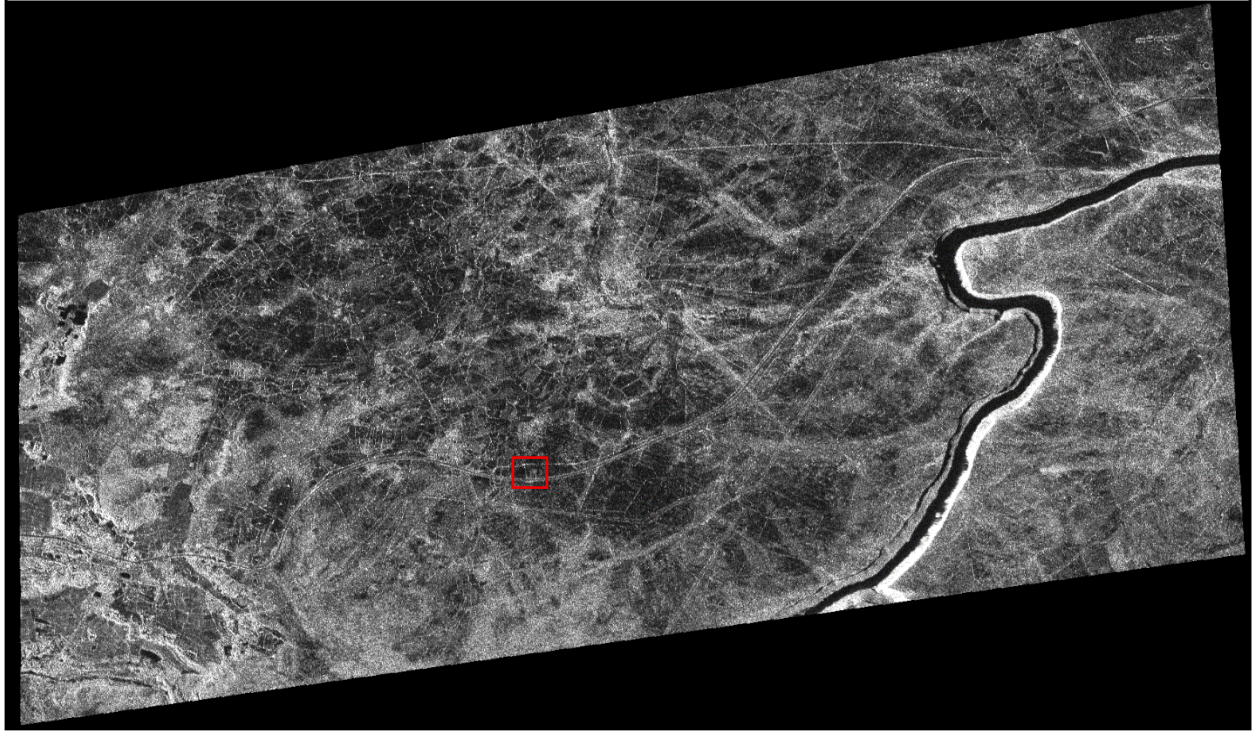
Flight Direction - DESCENDING

Polarization – VV+VH

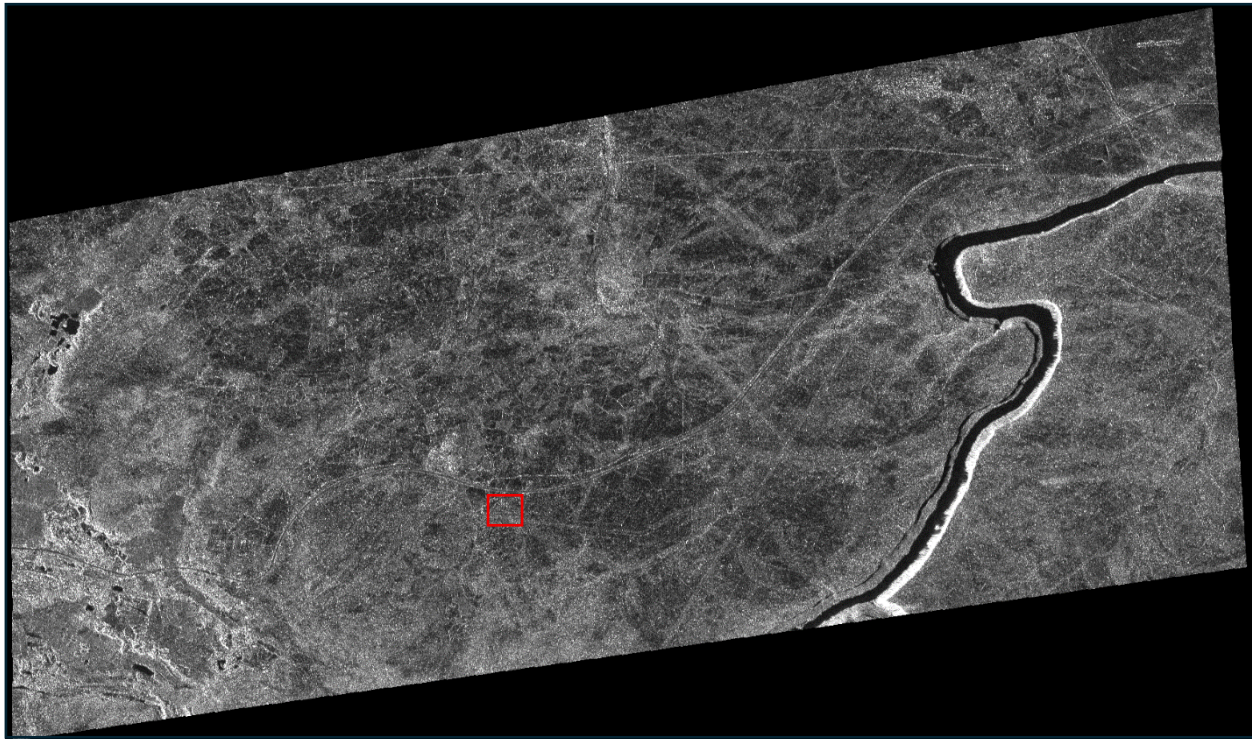
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

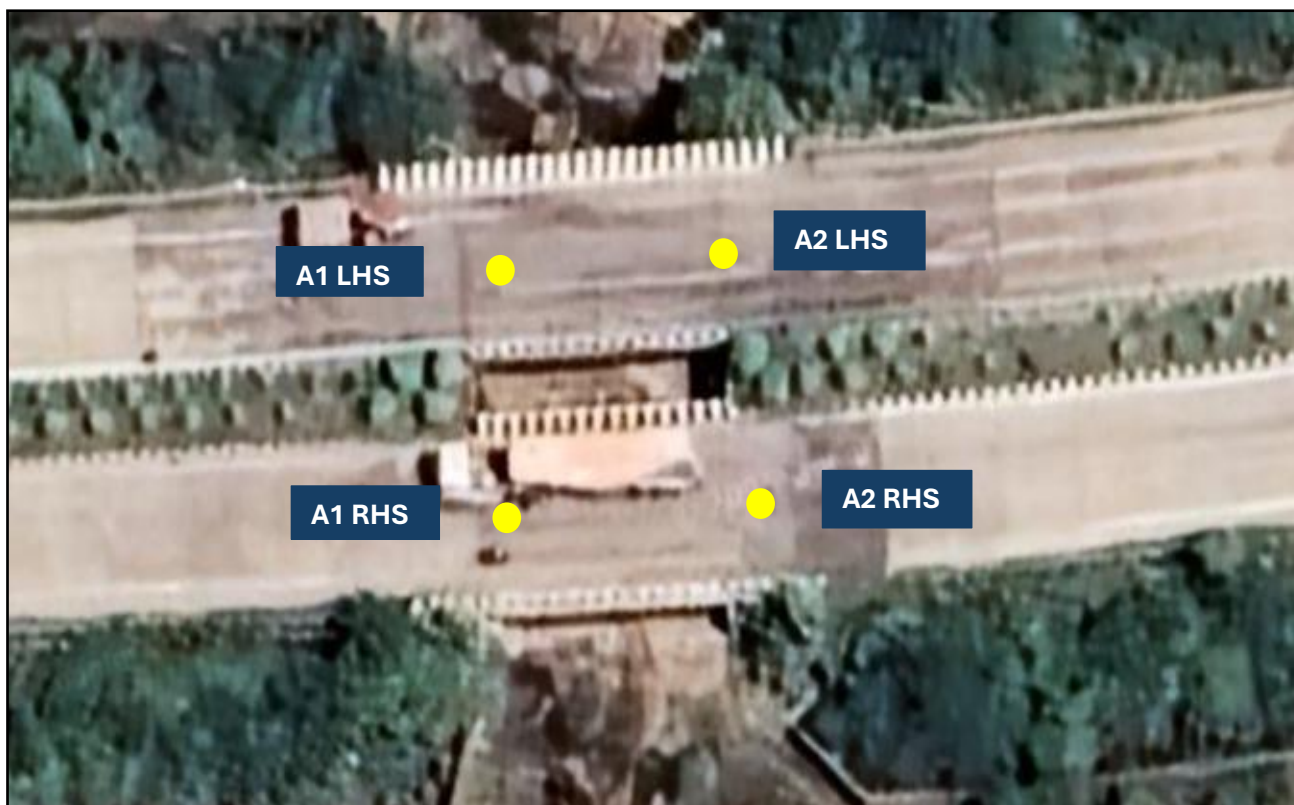


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

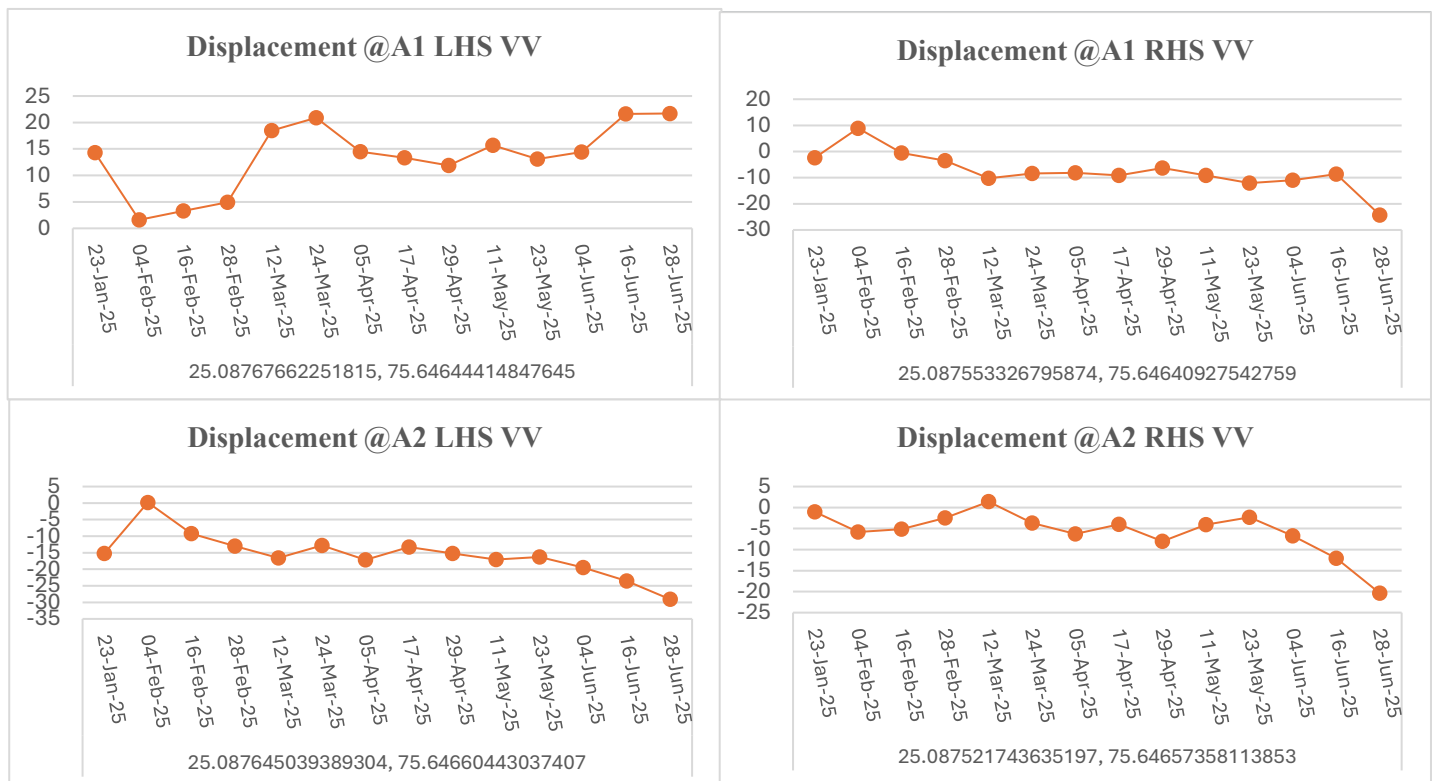
MNB – CH-1039+200

Length – 25.83M

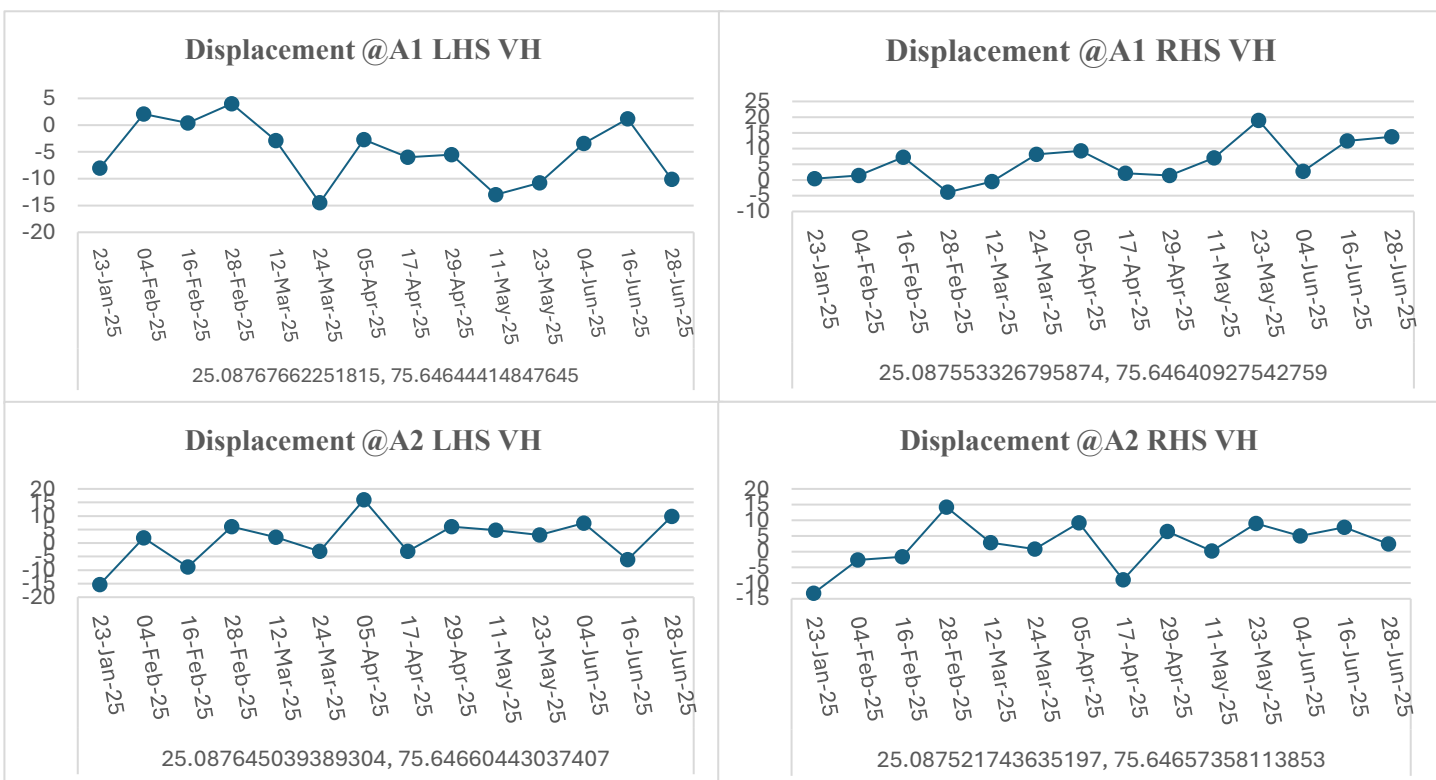


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



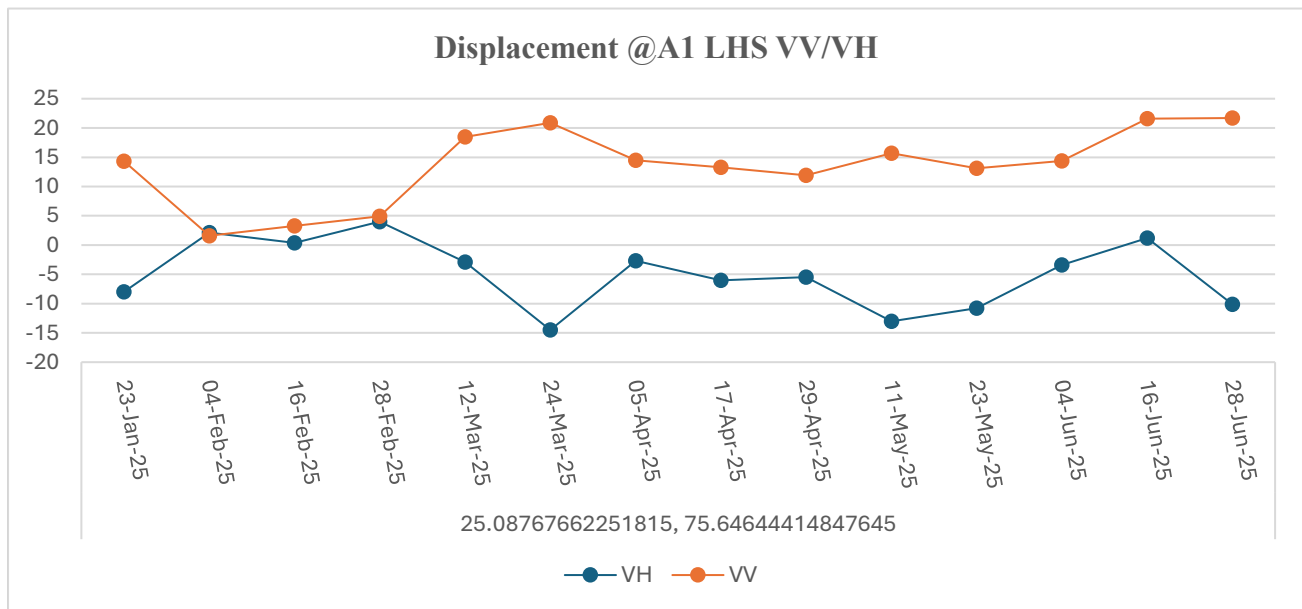
3.2 VH Displacement from 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.08767662251815, 75.64644414847645)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains predominantly positive throughout the monitoring period, indicating overall vertical uplift at the A1 LHS location.
- **Horizontal Displacement (VH)** Horizontal displacement shows persistent negative dominance with moderate variability, ranging roughly between -15 mm and +2 mm. VH values indicate continuous lateral settlement, with repeated negative excursions.

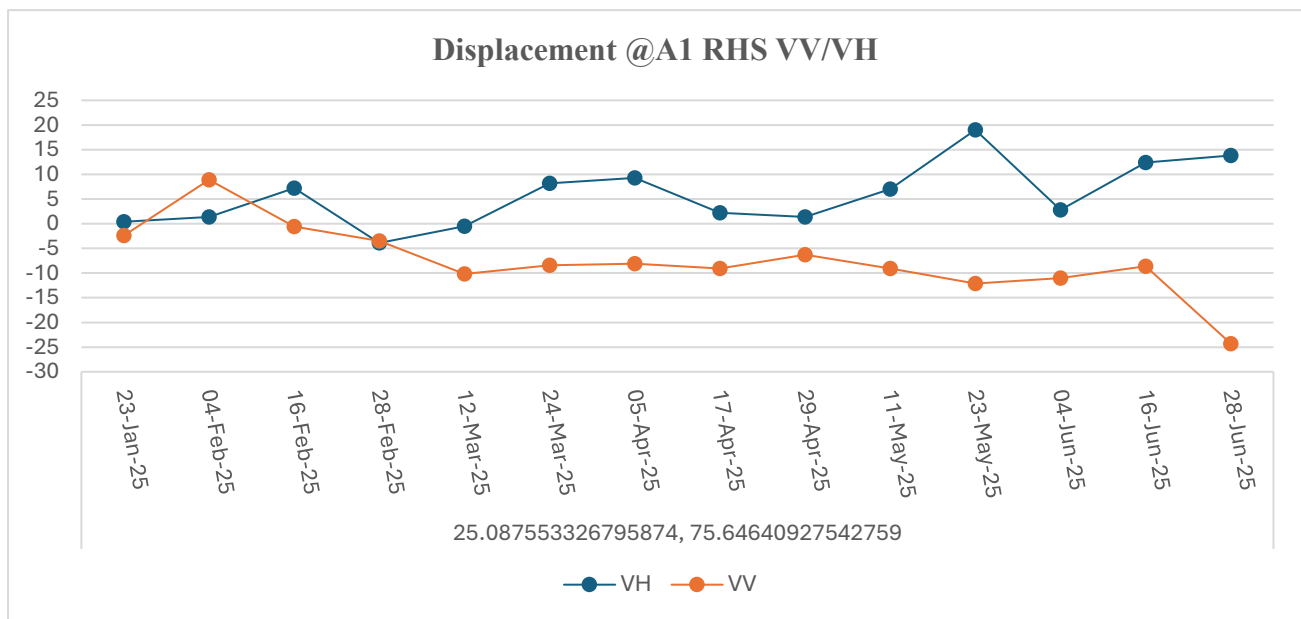
Critical Events & Anomalies:

- **February 2025:** VH briefly improves to 0 mm, while VV drops slightly to +2 mm, suggesting a short-lived lateral stabilization without loss of uplift.
- **May 2025:** VH drops again to -13 mm, while VV remains positive, representing another lateral settlement pulse without vertical collapse.
- **June 2025:** VH partially recovers to -3 mm, while VV remains high (+14-15 mm), suggesting temporary lateral relaxation under continued uplift.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.087553326795874, 75.64640927542759)



Key Observations:

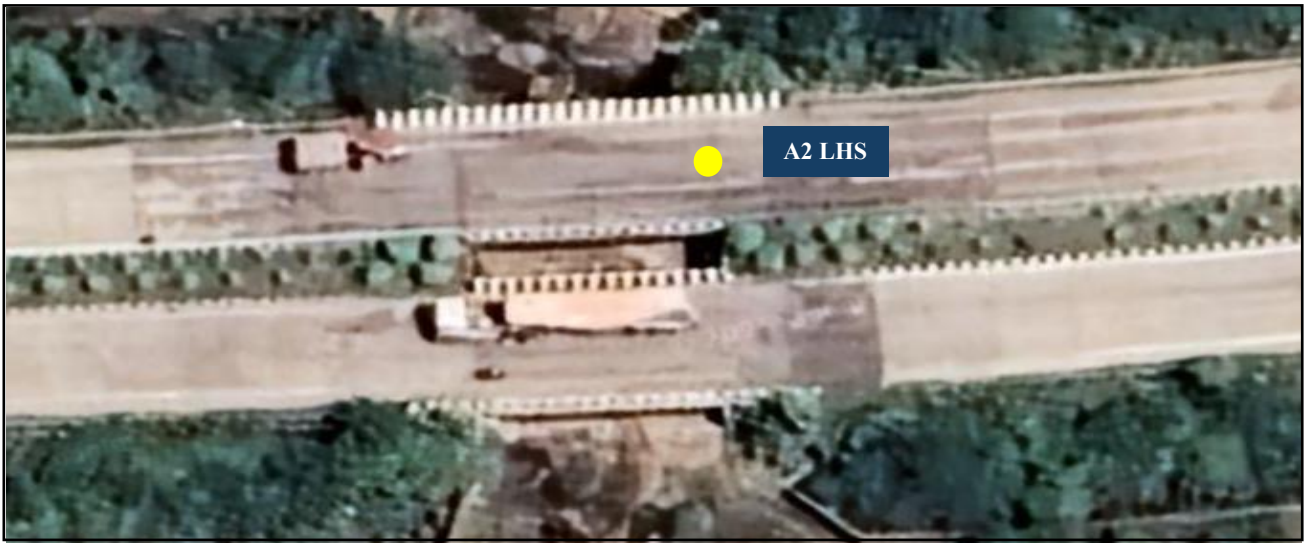
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows moderate variability with predominantly negative values, ranging roughly between +2 mm and –16 mm.
- **Vertical displacement (VV)** Vertical displacement shows a predominantly negative trend over the monitoring period (Jan–Jun 2025), indicating overall vertical settlement at the A1 RHS location. VV values range approximately from +9 mm to –26 mm.

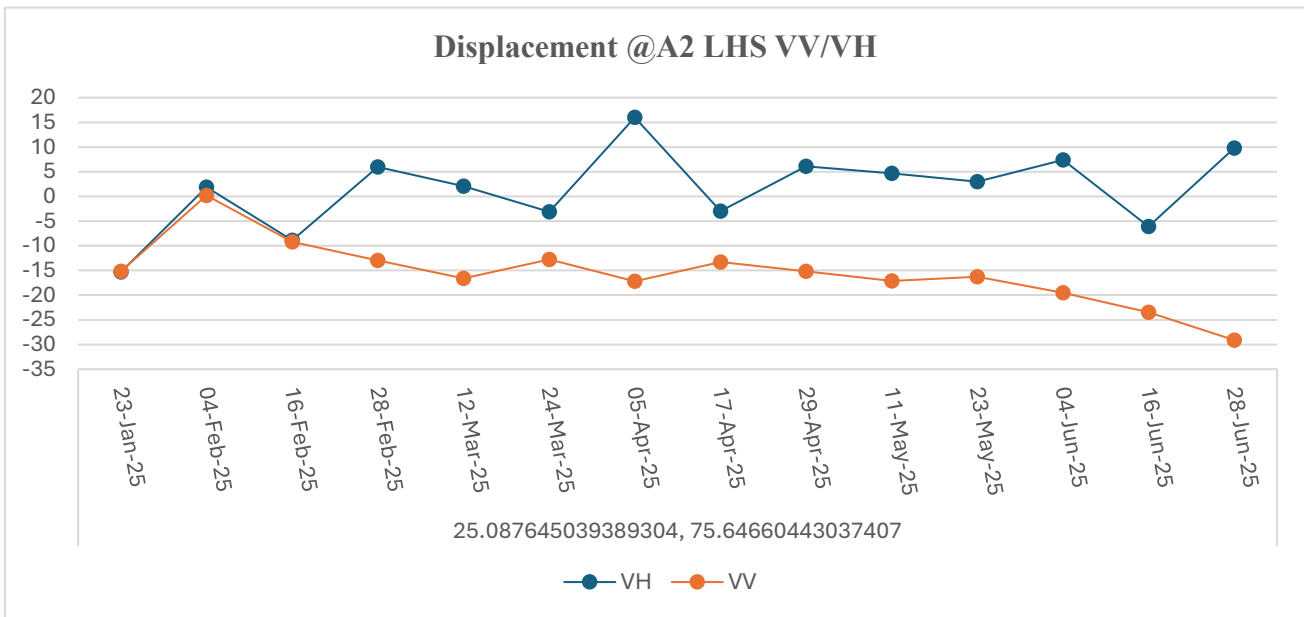
Critical Events & Anomalies:

- **February 2025:** VV reaches a local uplift peak (+9 mm) while VH increases to +2 mm, indicating a short-lived coupled uplift event affecting both vertical and horizontal components.
- **June 2025:** VH increases again to +12–14 mm, while VV reaches its maximum settlement (–26 mm), confirming continued deformation escalation toward the end of the monitoring period.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.087645039389304, 75.64660443037407)



Key Observations:

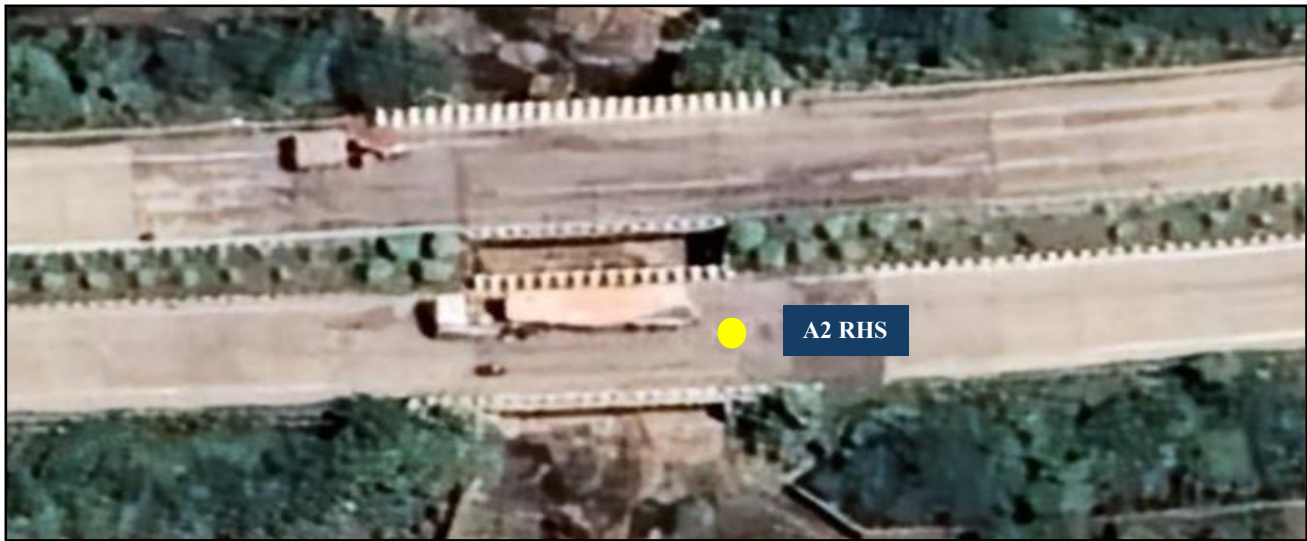
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits moderate variability with alternating positive and negative phases, ranging roughly between –8 mm and +15 mm.
- **Vertical Displacement (VV)** vertical displacement shows a strong and persistent negative trend throughout the monitoring period (Jan–Jun 2025), indicating progressive vertical settlement at the A2 LHS location.

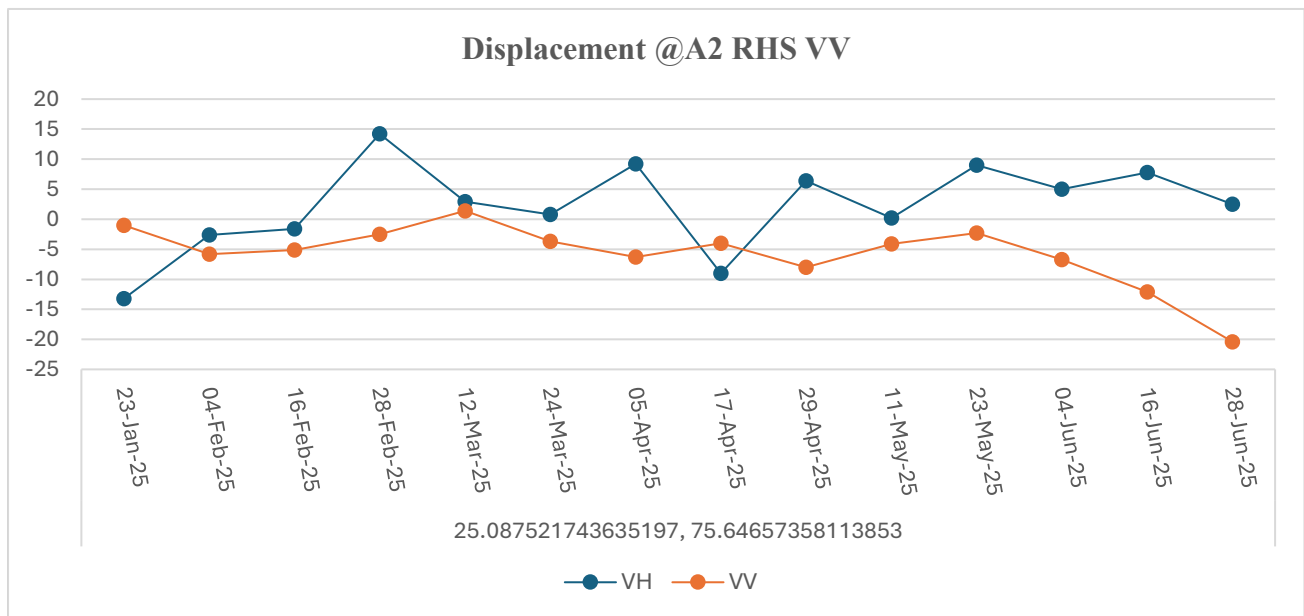
Critical Events & Anomalies:

- **February 2025:** VH briefly rises to +2 mm, while VV remains negative (–6 mm), indicating a short-lived lateral adjustment under ongoing vertical settlement.
- **May 2025:** VH fluctuates around +4 mm, while VV reaches –22 mm, indicating continued settlement dominance.
- **June 2025:** VH decreases briefly to –6 mm before recovering to +10 mm, while VV reaches its maximum settlement (–30 mm), confirming progressive deformation escalation toward the end of the monitoring period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.087521743635197, 75.64657358113853)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement exhibits a predominantly negative trend throughout the monitoring period, indicating overall vertical settlement at the A2 RHS location.
- **Horizontal (VH) displacement** Horizontal displacement shows moderate variability with predominantly positive values, ranging roughly between -14 mm and +14 mm.

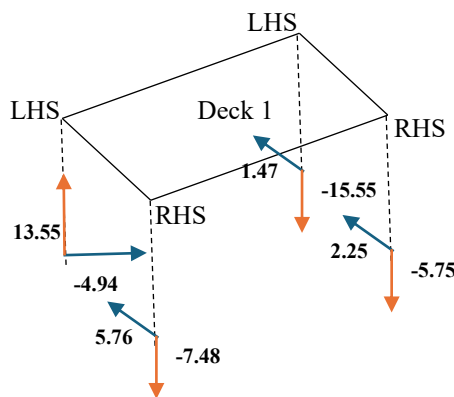
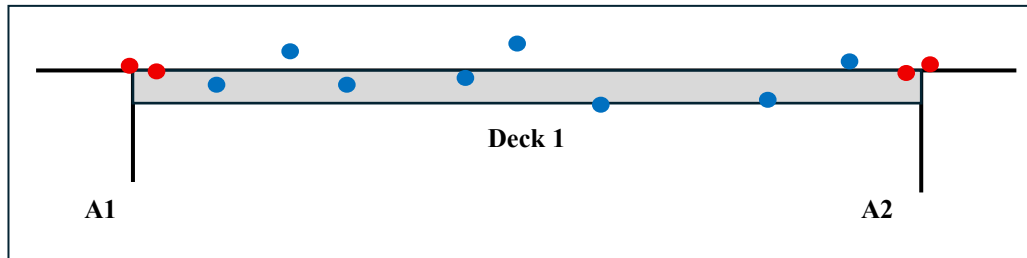
Critical Events & Anomalies:

- **January 2025:** VH begins at a negative value (-14 mm), while VV remains near 0 mm, indicating initial lateral settlement without significant vertical movement.
- **May 2025:** VH rebounds to +6-9 mm, while VV stabilizes around -4 to -6 mm, suggesting partial lateral recovery under continued settlement.
- **June 2025:** VV deteriorates sharply to -18 to -22 mm, while VH remains moderately positive (+3-8 mm), confirming accelerated settlement toward the end of the monitoring period without lateral stabilization.

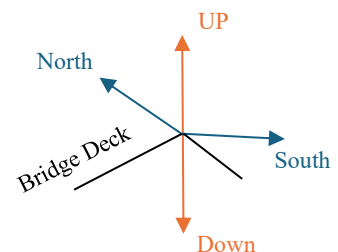
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits highly non-uniform vertical displacement, indicating pronounced differential vertical response between the LHS and RHS regions. The coexistence of uplift on the LHS and settlement on the RHS confirms vertical bending combined with torsional effects across Deck 1.

Horizontal Displacement Behaviour:

- Horizontal displacement shows clear directional variation across the deck width, reflecting combined translational and torsional behaviour. The maximum horizontal displacement is recorded near the central RHS region, with a value of approximately +2.25 mm, indicating lateral movement of the deck core.

CONCLUSION:

- The bridge deck slab demonstrates a complex deformation mechanism, characterized by simultaneous vertical uplift, settlement, and asymmetric horizontal displacement. The large uplift on the LHS combined with severe settlement on the RHS confirms strong torsional rotation coupled with vertical bending of the deck slab. Overall, the magnitude and distribution of displacements suggest structurally significant behaviour, warranting detailed assessment and continued monitoring, particularly under changing load or environmental conditions.